AIA Unit 5 Notes

1. Robotics — Introduction & Fundamentals ✗ [Not in PPT]

Robotics is an interdisciplinary field combining mechanical engineering, electrical engineering, computer science, and AI to design, build, program, and use robots. A robot is an autonomous or semi-autonomous machine that can sense its environment, process information, and take actions to accomplish a task.

Key Characteristics of a Robot:

- **Sensing:** Gathers data from the environment through sensors (cameras, LiDAR, touch, sound).
- **Processing:** Interprets sensor data and makes decisions using onboard computing and AI algorithms.
- **Actuation:** Physically interacts with the environment using motors, grippers, wheels, or arms.
- **Autonomy:** Operates without continuous human guidance — ranges from semi-autonomous to fully autonomous.

Types of Robots:

Type	Description	Example
Industrial Robots	Perform repetitive tasks in manufacturing — welding, painting, assembly.	Robotic arms in car factories
Mobile Robots	Navigate through environments autonomously.	Autonomous delivery robots, vacuum cleaners
Humanoid Robots	Human-like form and movement for interaction.	ASIMO, Boston Dynamics Atlas
Collaborative Robots (Cobots)	Work safely alongside humans on shared tasks.	Factory assembly assistance
Medical Robots	Assist in surgery and rehabilitation.	da Vinci surgical system
Autonomous Vehicles	Navigate roads using sensors and AI.	Tesla Autopilot, Waymo

Robotics and AI — The Connection

AI is the brain of modern robotics. Without AI, robots follow fixed, pre-programmed scripts. With AI, robots can perceive complex environments, learn from experience, make decisions under uncertainty, and adapt to new situations — this is what makes autonomous robotics possible.

2. Mobile Robot Hardware ✗ [Not in PPT]

A mobile robot's hardware is the physical body that enables it to move, sense, and interact. Key hardware components include:

Locomotion / Drive System ✗ [Not in PPT]

- **Differential Drive:** Two independently driven wheels — most common for ground robots. Turning is achieved by driving wheels at different speeds.
- **Omnidirectional Drive:** Mecanum or omni wheels allow movement in any direction without rotating the chassis.
- **Legged Robots:** Use legs for movement over rough terrain — more versatile but mechanically complex (e.g., quadrupeds, bipeds).
- **Tracked Drive:** Uses tank-like tracks — excellent for rough, uneven terrain (e.g., military robots, Mars rovers).

Onboard Computing ✗ [Not in PPT]

- **Microcontrollers:** Low-level control of motors and sensors (e.g., Arduino).
- **Single Board Computers:** Higher-level processing — running ROS (Robot Operating System), AI models (e.g., Raspberry Pi, NVIDIA Jetson).
- **GPUs:** Required for real-time deep learning inference — image recognition, obstacle detection.

Power System ✗ [Not in PPT]

- **Battery:** Lithium-ion or LiPo batteries are standard for mobile robots — balance energy density and weight.
- **Power Management:** Ensures stable power delivery to all components; monitors charge and protects from overload.

Actuators ✗ [Not in PPT]

- **DC / Servo Motors:** Drive wheels and joints — servo motors provide precise position control.
- **Pneumatic / Hydraulic Actuators:** Used in heavy industrial robots needing high force.
- **Grippers / End Effectors:** Interact with objects — range from simple two-finger grippers to complex multi-finger hands.

3. Path Planning for a Point Robot ✓ [In PPT]

Path planning involves calculating a collision-free, efficient route from a start to a goal location using a map of known obstacles. A point robot is a simplified model where the robot is treated as a single point in space — ignoring its physical dimensions.

Strategies:

- **Global Path Planning:** Plans the complete path from start to end in advance, assuming a static, fully known environment. Uses the full map.
- **Local Path Planning:** Reacts to changes in the environment in real-time, using sensor data to adjust the path as the robot moves. Handles dynamic obstacles.

Key Algorithms:

- **Visibility Graph:** Nodes represent start, goal, and obstacle vertices. Edges connect nodes with unobstructed straight lines. A* search finds the shortest path through the graph.
- **A* Search:** Uses f(n) = g(n) + h(n) to find the optimal path (covered in Unit IV).
- **Probabilistic Roadmap (PRM):** Randomly samples points in free space, connects nearby samples — used in high-dimensional spaces.
- **Rapidly-exploring Random Tree (RRT):** Grows a tree from the start by randomly sampling the space — efficient for complex environments.
- **Reinforcement Learning for Path Planning:** Can be used to learn optimal policies for path planning with deep RL, especially in unknown or dynamic environments.

General Steps for Path Planning:

1. Map Creation: Create or receive a map showing obstacles and free space.
2. Robot Model: Define the robot as a point (or with its actual dimensions via configuration space).
3. Path Generation: Apply a chosen algorithm (Visibility Graph, A*, RRT) to find a path.
4. Execution: Guide the robot along the computed path, using local planning to adjust for dynamic changes.

```
Visibility Graph Example:

  Start ──────────────────────────── Goal

            ┌───────┐
            │ Obst. │
            └───────┘

  Nodes: Start, Goal, 4 corners of obstacle
  Edges: All unobstructed straight-line connections
  A* finds shortest path: Start → corner1 → corner2 → Goal
```

4. Sensing and Mapping for a Point Robot (SLAM) ✓ [In PPT]

Sensing and mapping involve using sensors to gather environmental data and construct maps, enabling the robot to understand its surroundings and navigate through them using SLAM algorithms.

- **SLAM — Simultaneous Localisation and Mapping:** The process where a robot builds a map of an unknown environment while simultaneously keeping track of its own location within that map. It is the fundamental challenge of autonomous mobile robotics.

How SLAM Works:

1. Robot starts in an unknown environment with no prior map.
2. Sensors (LiDAR, cameras, IMU) collect data about surroundings.
3. SLAM algorithm processes sensor data to estimate robot position (localisation) and incrementally builds the map simultaneously.
4. As the robot moves, it matches new sensor readings against the existing partial map to refine both the map and its position estimate.

Common Sensors Used in SLAM:

- **LiDAR:** Provides accurate 2D or 3D point clouds — most common for SLAM. Creates occupancy grid maps.
- **Depth/Stereo Cameras:** Provide visual depth information for visual SLAM (vSLAM).
- **IMU (Inertial Measurement Unit):** Tracks the robot's own motion between sensor readings — compensates for odometry errors.
- **Odometry (Wheel Encoders):** Infer position from wheel rotation counts — cheap but accumulates drift over time.

Map Representations:

- **Occupancy Grid Maps:** Common in 2D LiDAR-based SLAM — grid where each cell stores probability of being occupied (obstacle) or free.
- **Geometric Maps:** Precise spatial coordinates for features and landmarks.
- **Topological Maps:** Graph of places and connecting paths — useful for high-level navigation planning.

Role of Sensing and Mapping:

- **Localisation:** Determining the robot's current position within the map.
- **Navigation:** Creating safe paths for the robot to follow.
- **Obstacle Avoidance:** Detecting and avoiding static and dynamic obstacles.
- **Object Recognition:** Identifying and classifying objects in the environment.

5. Contact Sensors ✓ [In PPT]

A contact sensor is a type of proximity sensor that detects the physical presence of an object by making actual physical contact with it. Unlike visual sensors (which use cameras), contact sensors require touching the object to detect it.

Key Properties:

- **Accuracy:** Detect objects and measure dimensions with high accuracy of up to 1 μm.
- **Structure:** Their strength to withstand sliding movement and slim bodies make them ideal for a wide variety of measurement applications.
- **Construction:** Consist of a sensor and a magnet. When attached to entry/exit points (windows, doors), they alert users when triggered.

How Contact Sensors Function:

1. When the sensor recognises separation (e.g., a door opening away from the magnet), it transmits a signal in real time.
2. This signal creates a notification in a security app or central monitoring station.
3. The signal can trigger a wide range of smart home functions depending on programmed rules.

Examples:

- **Door/Window Sensors:** Alert homeowners when a door or window is opened — core component of home security systems.
- **Refrigerator Sensor:** Notifies when a fridge door has been left open for more than a minute — reduces energy waste.
- **Industrial Contact Probes:** Measure precise dimensions of manufactured parts on production lines.

> **Contact vs Visual Sensors**
>
> Contact Sensors: require physical touch — high precision for dimensions, simple electronics. Visual Sensors (cameras): operate remotely — ideal when contact might damage the object or when spatial context is needed. Neither is universally better — the right choice depends on the application.

6. Inertial Sensors (IMU) ✓ [In PPT]

Inertial sensors are devices that measure changes in an object's linear and angular motion relative to an inertial reference frame. They are typically packaged as an Inertial Measurement Unit (IMU) — a single device combining multiple sensors, often on a single microchip using MEMS technology.

- **MEMS Technology:** Micro Electro Mechanical Systems — microscale mechanical structures that detect minute changes in position, velocity, and orientation. Enable tiny, low-power inertial sensors in consumer devices.

Three Components of an IMU:

Component	What it Measures	How it Works
Accelerometer (3-axis)	Linear acceleration along X, Y, Z axes (including gravity)	A tiny internal mass lags behind due to inertia when the sensor accelerates — the displacement is measured electrically.
Gyroscope (3-axis)	Rotational motion / angular rate around X, Y, Z axes	Measures the Coriolis effect on a vibrating MEMS structure when it rotates.
Magnetometer (optional)	Earth's magnetic field — provides absolute heading (like a compass)	Detects magnetic field strength and direction to give an absolute north reference — corrects gyroscope drift.

Applications of Inertial Sensors:

- **Automotive:** Safety systems — airbag deployment (detects sudden deceleration), electronic stability control, driver assistance.
- **Robotics & Autonomous Navigation:** Determines a robot's position and orientation for navigation and movement control — essential for SLAM.
- **Wearable Technology:** Tracking player movements in sports, monitoring physical activity, detailed motion analysis.
- **Smartphones:** Screen rotation, step counting, gaming orientation.
- **Aerospace:** Inertial navigation systems (INS) for aircraft and missiles.

IMU Drift Problem

Gyroscopes and accelerometers accumulate small errors over time (called drift) — a tiny measurement error compounds with each integration step. Magnetometers provide an absolute reference to correct gyroscope drift, and SLAM algorithms fuse IMU data with sensor data (like LiDAR) to correct accumulated errors.

7. Infrared (IR) Sensors ✓ [In PPT]

An infrared sensor (IR sensor) is a radiation-sensitive optoelectronic component with spectral sensitivity in the infrared wavelength range (780 nm to 50 μm). IR sensors detect heat radiation (infrared radiation) that changes over time and space due to movement of people or objects.

IR sensors are now widely used in motion detectors — used in building services to switch on lamps or in alarm systems to detect intruders. All objects radiate some form of thermal radiation in the infrared spectrum — these are invisible to the human eye but detectable by IR sensors.

Two Main Components:

- **IR LED (Emitter):** Emits infrared light at a specific wavelength.
- **IR Photodiode (Detector):** Sensitive to IR light of the same wavelength emitted by the IR LED. When IR light falls on it, resistance and output voltage change proportionally to the intensity of IR light received.

Five Basic Elements of an IR Detection System:

1. Infrared source (IR LED or laser)
2. Transmission medium (air, fibre optic)
3. Optical components (lenses to focus IR beam)
4. IR detector / receiver (photodiode)
5. Signal processing (amplification, thresholding, output)

Types of IR Sensors:

- **Active IR Sensor:** Emits its own IR beam and detects the reflection from nearby objects. Used for proximity detection.
- **Passive IR Sensor (PIR):** Detects IR radiation emitted by warm objects (humans, animals) in its field of view — does NOT emit. Used in motion detection alarms and automatic lighting.

Application	Type Used	Description
Motion Detection (Security)	PIR	Detects body heat from intruders in alarm systems and automatic lighting.
Proximity Detection	Active IR	Detects nearby objects without physical contact — e.g., smartphone screen turning off near face.
Obstacle Avoidance (Robots)	Active IR	Short-range detection of obstacles for robot navigation.
Remote Controls	Active IR	TV remotes transmit IR signals to control devices.
Temperature Measurement	Passive IR	Non-contact temperature sensing in thermometers.

8. SONAR (Sound Navigation and Ranging) ✓ [In PPT]

Sonar is a technology that uses sound waves — particularly ultrasonic waves — to detect and measure distances to objects, or to map an area. It works on the principle of echo: a pulse of sound is sent out and the time taken for the reflected echo to return is measured to determine the object's distance, direction, and speed.

How Sonar Works — Step by Step:

1. Sound Emission: An active sonar system sends out a pulse of high-frequency (ultrasonic) sound into the medium (water or air).
2. Reflection: When sound waves encounter an object, they bounce off it, creating an echo.
3. Echo Detection: A receiver on the sonar system detects these reflected echoes.
4. Calculation: By measuring the time taken for the echo to return, the system calculates the distance to the object.

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Distance Calculation:
  Distance = (Speed of Sound × Time for round trip) / 2
```

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Speed of sound in water ≈ 1500 m/s
Speed of sound in air   ≈ 343 m/s
(Divide by 2 because sound travels TO the object AND back)
```

Types of Sonar:

- **Active Sonar:** Emits its own acoustic pulses and listens for reflected echoes. Detects objects but reveals the sonar's own position.
- **Passive Sonar:** Detects existing sounds in the environment (e.g., engine noise from a submarine) without emitting any signal. Covert — does not reveal its own presence.

Application	Description
Ocean Exploration & Mapping	Creates nautical charts, maps the seafloor, and explores underwater landscapes.
Fishing	Locates schools of fish and maps their distribution, density, and movement.
Navigation	Detects underwater hazards such as icebergs and submerged rocks.
Defence	Detecting submarines, mine hunting, and underwater surveillance.
Robotics (Ultrasonic)	Short-range obstacle detection in robots and autonomous vehicles.

9. RADAR (Radio Detection and Ranging) ✓ [In PPT]

RADAR is an electromagnetic sensor used for detecting, locating, tracking, and recognising objects at considerable distances. It operates by transmitting electromagnetic energy toward targets and observing the echoes returned from them.

- Targets can be aircraft, ships, spacecraft, vehicles, astronomical bodies, birds, insects, or even rain.
- Can determine presence, location, velocity, and sometimes size and shape of targets.
- Key advantage: detects faraway objects under adverse weather conditions and determines range with precision.
- Active sensing device — has its own source of illumination (transmitter).
- Operates in the microwave region: approximately 400 MHz to 40 GHz.

Fundamentals of RADAR — How it Works:

1. Transmission: A narrow beam of electromagnetic energy is radiated from an antenna into space.
2. Target Illumination: When the beam hits a target, it intercepts some energy and reflects a portion back.
3. Reception: A receiver (often sharing the same antenna on a time-shared basis) extracts the reflected echo signals.

4. Range Measurement: Total round-trip time for the signal × speed of light / 2 = distance to target.
5. Direction: Angular direction determined by the direction the antenna points when the echo is received.
6. Track: Measuring target location at successive instants builds the target's track and predicts future path.

Application	Description
Air Traffic Control	Tracking positions of aircraft in real time to manage safe separation.
Weather Observation	Detecting rain, storms, and severe weather events.
Navigation	Aircraft and ship navigation — especially in poor visibility.
Speed Measurement	Law enforcement speed guns and industrial process control.
Space Surveillance	Tracking satellites and debris in orbit.
Military	Target acquisition, missile guidance, battlefield surveillance.

RADAR vs SONAR — Key Comparison

RADAR: uses electromagnetic waves (radio/microwave) — works in air/space, long range, affected by metal objects. SONAR: uses sound waves — works best in water (poor radio propagation underwater), shorter range in air. Both use the echo principle: transmit a pulse, measure time for echo to return, calculate distance.

10. Laser Rangefinders ✓ [In PPT]

Laser rangefinders work by emitting a short laser pulse that travels to a target, reflects off its surface, and returns to the device. The rangefinder precisely measures the time taken for the round trip and, using the constant speed of light, calculates the distance to the target. This is called the Time-of-Flight (ToF) principle.

```
Time-of-Flight Distance Calculation:

  Distance = (Speed of Light × Time for round trip) / 2

  Speed of light = 3 × 10^8 m/s
  Example: echo returns in 10 ns
  Distance = (3×10^8 × 10×10^-9) / 2 = 1.5 metres
```

Key Components:

- **Laser Emitter:** Generates and sends a short, focused pulse of laser light toward the target.
- **Receiver / Detector:** A sensitive photodetector that captures the reflected laser pulse.
- **Timer:** A highly accurate internal timer measures the exact time difference between emission and reception.

- **Optics:** Lenses that focus the laser beam and collect the reflected light — often includes magnification for distant targets.
- **Processing Unit:** Converts the measured time into a distance value.

Application	Description
Military & Defence	Weapon targeting systems, artillery rangefinding, surveillance.
Construction & Surveying	Precise distance and area measurements in building, architecture, and land surveying.
Robotics (LiDAR)	A laser rangefinder swept in 2D or 3D creates a LiDAR point cloud for SLAM and obstacle detection.
Sports & Hunting	Golfers, hunters, and athletes measure distances to targets.
Autonomous Vehicles	LiDAR (rotating laser rangefinder) maps surroundings at high speed for navigation.

Laser Rangefinder vs LiDAR

A laser rangefinder measures distance to ONE point at a time. LiDAR (Light Detection and Ranging) is essentially a laser rangefinder that rotates rapidly — firing laser pulses in many directions per second to build a full 2D or 3D point cloud map of the surroundings. LiDAR is the sensor version; the laser rangefinder is the core measurement principle.

11. Biological Sensing ✗ [Not in PPT]

Biological sensing refers to sensors inspired by or directly derived from biological organisms — mimicking the sensory mechanisms found in living creatures to create more effective, energy-efficient sensing systems.

Types of Biologically Inspired Sensors:

- **Artificial Compound Eyes:** Inspired by insect eyes — wide field of view, good at detecting motion. Used in surveillance and robotics for panoramic vision.
- **Electronic Nose (e-Nose):** Mimics the mammalian olfactory system — arrays of chemical sensors detect volatile compounds. Used in food quality, medical diagnosis, environmental monitoring.
- **Electronic Skin (e-Skin):** Flexible sensor arrays that mimic human skin's ability to detect pressure, temperature, and texture. Used in prosthetics and humanoid robots.
- **Cochlear-Inspired Sensors:** Mimic the human ear's frequency analysis. Used in audio processing and speech recognition hardware.
- **Lateral Line Sensors:** Inspired by the fish lateral line system — detect water flow and pressure waves. Used in underwater robots for hydrodynamic sensing.

Advantages of Biologically Inspired Sensing:

- **Energy Efficiency:** Biological sensors achieve remarkable performance with minimal energy — a key inspiration for low-power robotics.

- **Robustness:** Evolved over millions of years to handle real-world noise and variability.
- **Parallel Processing:** Many biological sensors process information in parallel — a model for neuromorphic computing.

12. Robot Control System — Vertical Decomposition ✓ [In PPT]

Robot control system decomposition divides complex robotic tasks into manageable parts for more robust and flexible control. Vertical decomposition structures the control system hierarchically, with layers representing increasing levels of abstraction.

Hierarchical Structure:

- **Higher layers:** Handle complex behaviours, high-level decision-making, goals, and planning.
- **Lower layers:** Manage simpler, fundamental tasks — motor control, reflex responses, sensor reading.
- **Key Principle:** Each layer builds upon the one below it — abstract goals at the top, physical actions at the bottom.

Task-Achieving Behaviours:

A key approach in vertical decomposition is to break the robot's functionality into 'task-achieving behaviours' — each behaviour is a functional unit that operates at a specific level of competence, from basic obstacle avoidance (low level) to complex path navigation (high level).

```
Vertical Decomposition — Layer Stack (top to bottom):

  Level 4: Mission Planning      ← High abstraction (goals, strategy)
  Level 3: Path Planning         ← Navigate from A to B
  Level 2: Obstacle Avoidance    ← React to immediate threats
  Level 1: Motor Control         ← Low-level wheel/joint commands
  Level 0: Sensor Interface      ← Raw sensor reading
```

13. Robot Control System — Horizontal Decomposition ✓ [In PPT]

Horizontal decomposition breaks the robot control system into specialised functional units (modules) that operate in parallel, each responsible for specific functions such as perception, modelling, path planning, and motor control.

Key Concepts:

- **Functional Units:** Specialised modules — each does one job well (e.g., perception module, planning module, motor control module).
- **Parallel Operation:** Functional units operate independently and simultaneously — sensor processing continues while the planner computes a new path.

Blackboard Architecture:

- **Blackboard:** A shared database that acts as the central communication hub — all modules read from and write to it.
- **Knowledge Sources:** Specialised reasoning agents that contribute their expertise to the blackboard when triggered.
- **Control Unit (Monitor & Scheduler):** Observes the blackboard and triggers the appropriate knowledge sources when their specialised tasks are needed.

```
Horizontal Decomposition — Blackboard Model:

┌─────────────────────────────────────────────┐
│              BLACKBOARD (shared data)        │
└─────────────────────────────────────────────┘
           │                      │
           ▼                      ▼
┌──────────────────┐   ┌──────────────────────┐
│ Perception Module│   │  Planning Module      │
│ (reads sensors)  │   │ (computes path)       │
└──────────────────┘   └──────────────────────┘
           │                      │
           ▼                      ▼
┌─────────────────────────────────────────────┐
│            Motor Control Module              │
│      (executes movement commands)            │
└─────────────────────────────────────────────┘
```

How Vertical and Horizontal Work Together:

These decomposition methods are commonly combined. Rodney Brooks's influential subsumption architecture for mobile robots uses both: a vertical hierarchy of behavioural layers, where each layer contains horizontally organised functional units for perception, planning, and motor control.

- **Benefit:** More robust, flexible, and testable robot control systems — complex problems are broken into simpler, manageable parts that can be developed and tested independently.

14. Middleware ✗ [Not in PPT]

Middleware in robotics is a software layer that sits between the robot's hardware (sensors, actuators) and the high-level AI/application software. It provides standardised services and communication infrastructure that allows different hardware and software components to work together seamlessly.

What Middleware Does:

- **Hardware Abstraction:** Hides the complexity of specific hardware — a camera, for example, appears the same to the application layer regardless of its brand or interface.
- **Inter-Process Communication:** Manages message passing between different software modules (e.g., the perception module sending detected objects to the planning module).
- **Timing and Synchronisation:** Ensures sensor data from different sources (camera, LiDAR, IMU) is properly time-stamped and synchronised.
- **Reusability:** Standardised interfaces mean software modules written by different teams can be combined.

ROS — Robot Operating System:

ROS (Robot Operating System) is the most widely used robotics middleware. Despite the name, it is not an operating system — it is a middleware framework running on top of Linux.

- **Nodes:** Individual software processes (e.g., camera_driver, object_detector, path_planner).
- **Topics:** Named message channels — nodes publish to and subscribe from topics.
- **Services:** Synchronous request-response communication between nodes.
- **ROS4HRI:** A standardised ROS extension for Human-Robot Interaction — enables sharing of HRI algorithms across different robot platforms.

Why Middleware Matters

Without middleware, every robotics project must rebuild communication infrastructure from scratch — a huge waste of effort. ROS allows researchers and engineers worldwide to share and reuse robot software components, accelerating the entire field.

15. High-Level Control ✗ [Not in PPT]

High-level control in robotics refers to the planning, reasoning, and decision-making layers that operate above the immediate sensor-actuator loop. While low-level control manages motor speeds and sensor readings, high-level control answers the question: 'What should the robot do and why?'

Components of High-Level Control:

- **Mission Planning:** Defines the overall goal or task sequence the robot must complete. E.g., 'Inspect all shelves in this warehouse.'
- **Task Allocation:** In multi-robot systems, decides which robot handles which sub-task.
- **Behaviour Management:** Activates and deactivates lower-level behaviours based on the current situation and mission requirements.
- **Deliberative Planning:** Uses AI planning techniques (STRIPS, POP — covered in Unit IV) to reason about action sequences and their effects.
- **World Model Maintenance:** Maintains an up-to-date internal model of the environment — updated continuously from sensor data.

Relationship Between Control Levels:

```
High-Level Control   ← Mission goals, deliberative planning
       ↓  (goal decomposition)
Mid-Level Control    ← Path planning, behaviour management
       ↓  (motion commands)
Low-Level Control    ← Motor control, PID loops, sensor reading
       ↓  (physical actions)
Robot Hardware       ← Wheels, joints, sensors
```

16. Human-Robot Interaction (HRI) ✓ [In PPT]

Human-Robot Interaction (HRI) is the study and design of systems enabling efficient, safe, and natural interactions between humans and robots. It is a multidisciplinary field drawing contributions from AI, robotics, natural language processing, human-computer interaction, design, psychology, and philosophy.

What HRI Aims to Achieve:

- **User Understanding:** Improve human comprehension of the robot's state, actions, and environment — enhancing situational awareness.
- **Intuitive Control:** Allow users to command robots naturally using gestures, voice commands, eye gaze, or body movements — no programming knowledge required.
- **Collaboration:** Develop collaborative robots (cobots) that can work safely and effectively alongside humans on shared tasks.
- **Adaptability:** Build robots that understand human behaviour, adapt to changing environments, and respond to dynamic situations.

Physical HRI (pHRI) ✓ [In PPT]

A subfield of HRI focused on device design to enable people to safely interact physically with robotic systems. Concerns include force control, collision detection, compliant mechanisms, and safety standards.

Key Aspects of HRI:

Aspect	Description
User Understanding	HRI focuses on how humans can understand the robot's perspective, actions, and status — improving situational awareness and trust.
Intuitive Control	Interfaces designed to allow natural commands: gestures, voice, body movements — no specialist training needed.
Collaboration	Cobots work safely alongside humans; designed to detect changes in force and adjust movements for precision tasks.
Adaptability	Robots built to understand human behaviour and adapt to changing environments and dynamic situations.
Multidisciplinary	Draws on computer science, AI, engineering, psychology, design, and philosophy.

Examples of HRI Applications:

- **Industrial Settings:** Robots work with humans in factories — systems detect changes in force or adjust for precision assembly tasks alongside workers.
- **Retail:** HRI helps robots provide customer navigation, personalised product guidance, and efficient transaction processing.
- **Social Robots:** Humanoid robots mimic human communication and social interaction — suitable for public environments, customer service, eldercare.

Future Directions in HRI:

- **Enhanced Transparency:** Developing architectures like the Human-Swarm-Teaming Transparency and Trust Architecture (HST3) to improve transparency and build trust in autonomous systems.
- **Standardisation:** Creating standardised interfaces such as ROS4HRI to enable better sharing and comparison of HRI algorithms across platforms.
- **Emotional Intelligence:** Designing robots with greater awareness of human emotional states for a more positive, empathetic interaction experience.

17. Hybrid Control Architecture ✓ [In PPT]

A hybrid control architecture integrates different control strategies to leverage their respective strengths — most commonly combining a deliberative (planning) system and a reactive (real-time response) system to manage complex, dynamic environments.

Core Idea:

Deliberative systems are great at careful, optimal planning but are too slow for sudden real-world surprises. Reactive systems respond instantly but cannot plan ahead. A hybrid architecture uses both: plan carefully when time permits, react instantly when the environment demands it.

Key Layers:

Layer	Role	Time Scale
Deliberative Layer	Handles long-term goals, creates complex plans, operates on a global model of the environment.	Seconds to minutes
Reactive Layer	Provides immediate responses to unexpected local changes — obstacle avoidance, collision prevention.	Milliseconds
Execution / Transition Layer	Bridges deliberative and reactive layers — translates high-level plans into executable low-level commands, coordinates modules.	Milliseconds to seconds

Additional Components:

- **Sensors & Actuators:** Gather environmental information and translate control decisions into physical actions.
- **Communication Network:** (e.g., CAN bus) Interconnects control units, enabling exchange of data and commands between modules.

How Hybrid Control Works — Step by Step:

1. Sensing: Sensors collect data about the environment.
2. Deliberation: The deliberative layer uses sensor data and its internal model to plan a safe, optimal path toward the goal.
3. Normal Execution: Robot follows the planned path, executing the sequence of actions.

4. Reactive Override: If an unexpected obstacle or emergency appears, the reactive layer immediately takes over — generating a real-time evasive manoeuvre without waiting for the planner.
5. Resumption: Once the immediate threat is handled, control passes back to the deliberative layer to re-plan if needed.

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Hybrid Architecture — Control Flow:

  Mission Goal
       ↓
  [Deliberative Layer] — plans optimal path using global map
       ↓  (planned action sequence)
  [Execution Layer]    — translates plan to motor commands
       ↓
  [Reactive Layer]  ←  sensor detects sudden obstacle
       ↓             (immediately overrides planner)
  Robot avoids obstacle → [Reactive Layer] releases control
       ↓
  [Deliberative Layer] re-plans remaining path
```

Applications:

- **Autonomous Robotics:** Navigation in dynamic or partially known environments — warehouse robots, delivery drones, autonomous vehicles.
- **Hybrid Electric Vehicles:** Manages power distribution between the combustion engine and electric motor for optimal efficiency under varying conditions.

Why Hybrid Architecture Became the Standard

Early AI robots used only deliberative control — they were too slow to react safely. Early reactive robots (like Brooks' subsumption architecture) responded fast but couldn't plan. Hybrid architectures solved this by letting each layer do what it is best at — plan when time allows, react when safety demands.

QUICK REVISION — Unit V

All Topics — PPT Status & One-Line Summary ✓ [In PPT]

Topic	In PPT?	One-Line Summary
Robotics Fundamentals	✗ Not in PPT	Interdisciplinary field: sense + process + actuate. Types: industrial, mobile, humanoid, cobot.
Mobile Robot Hardware	✗ Not in PPT	Drive systems, onboard computing, power, actuators — the physical body enabling movement.
Path Planning for Point Robot	✓ In PPT	Calculate collision-free route start→goal. Global (full map) vs Local (real-time). Visibility Graph + A*.
Sensing & Mapping (SLAM)	✓ In PPT	Robot builds map + tracks own location simultaneously. Sensors: LiDAR, cameras, IMU, odometry.
Contact Sensors	✓ In PPT	Detect objects by physical touch. Accuracy up to 1 μm. Used in security/smart home systems.
Inertial Sensors (IMU)	✓ In PPT	Accelerometer (linear accel) + Gyroscope (rotation) + Magnetometer (heading). MEMS technology.
Infrared (IR) Sensors	✓ In PPT	Detect heat radiation. Active IR (proximity) vs Passive IR/PIR (motion detection).
SONAR	✓ In PPT	Uses sound echoes to detect objects. Active (emits pulses) vs Passive (listens). Used in ocean, defence, robotics.
RADAR	✓ In PPT	Uses radio/microwave echoes. Detects range, direction, velocity. Works in adverse weather. 400 MHz–40 GHz.
Laser Rangefinders	✓ In PPT	Time-of-Flight: laser pulse travel time → distance. Foundation of LiDAR.
Biological Sensing	✗ Not in PPT	Sensors inspired by biology: e-nose, e-skin, compound eyes, lateral line. Energy-efficient, robust.
Vertical Decomposition	✓ In PPT	Hierarchical layers: high-level planning → mid-level navigation → low-level motor control.
Horizontal Decomposition	✓ In PPT	Parallel functional modules + Blackboard architecture. Modules communicate via shared data store.
Middleware (ROS)	✗ Not in PPT	Software layer between hardware and AI. ROS: nodes communicate via topics. ROS4HRI for HRI.

Topic	In PPT?	One-Line Summary
High-Level Control	✗ Not in PPT	Mission planning, task allocation, deliberative planning — the 'what and why' of robot behaviour.
Human-Robot Interaction (HRI)	✓ In PPT	Study/design of human-robot systems. Goals: intuitive control, collaboration, adaptability.
Hybrid Control Architecture	✓ In PPT	Combines deliberative (planning) + reactive (real-time). Each layer does what it does best.

Sensor Comparison — Master Table ✓ [In PPT]

Sensor	Signal Used	Range	Key Use in Robotics	In PPT?
Contact Sensor	Physical contact	~0 mm (touch)	Detect object presence, measure dimensions	✓
IMU	Inertial forces	N/A (self-motion)	Track robot's own motion, orientation for SLAM	✓
IR Sensor	Infrared light	cm to metres	Short-range obstacle detection, motion detection	✓
SONAR	Ultrasonic sound	cm to tens of metres	Obstacle avoidance, underwater mapping	✓
RADAR	Radio / Microwave	metres to km	Long-range detection, velocity measurement	✓
Laser Rangefinder / LiDAR	Laser light (ToF)	cm to hundreds of metres	SLAM, 3D mapping, autonomous navigation	✓
Camera	Visible light	cm to km	Visual perception, object recognition, vSLAM	✓ (context)
Biological Sensors	Varies (chemical, pressure, flow)	Varies	Smell, skin sensation, flow detection	✗